**PROJECT REPORT**

# NurtureAI

Child Health, Behaviour, and Parenting Support Platform

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# Abstract

NurtureAI is a comprehensive, AI-powered web platform designed to bridge the gap between clinical data science and practical healthcare guidance in the parenting and paediatric domains. The system integrates four specialised machine learning models and a RAG-powered chatbot module, each addressing a distinct health or behavioural challenge: adult behaviour and parenting risk analysis, celiac disease screening, infant mortality risk assessment, and problematic internet use prediction in older children.

Built on a Flask-based RESTful backend with a responsive vanilla JavaScript Single Page Application (SPA) frontend, NurtureAI delivers real-time predictive analytics through a wizard-style interface accessible to non-technical parents. The platform incorporates a Retrieval-Augmented Generation (RAG) chatbot powered by ChromaDB (16,407 MedQuAD Q&A pairs) and Google Gemma 4 31B, enabling contextually grounded medical Q&A with emergency triage escalation flags.

EDA conducted across all core datasets revealed right-skewed SII distributions in CMI-PIU data, strong vaccination-mortality correlations in NFHS-5 records, and high-discriminating symptom co-occurrence patterns in celiac data. The integrated platform demonstrates that child health risk assessment is a measurable, structured, and statistically observable multi-domain problem amenable to a unified AI-powered parenting companion application.

# Declaration

We, the undersigned students of B.Tech (CSE) – AI and Data Engineering, hereby declare that this project report, titled "NurtureAI", is an authentic record of the research and development work carried out by us during the academic semester. The dataset collection, machine learning model development, exploratory data analysis, full-stack integration, and subsequent interpretation are the result of our own efforts.

All sources of information, datasets, and libraries used in this project have been duly acknowledged. We confirm that this work has not been submitted in part or in full for any other diploma or degree.

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# 1. Introduction & Problem Formulation

## 1.1 Background & Motivation

The domain of child health and parenting encompasses a wide range of interrelated challenges — from behavioural and mental health risks in children to life-threatening conditions such as celiac disease and infant mortality. These challenges are complex, multi-factorial, and data-rich, making them well-suited to statistical and machine learning analysis.

Parents and caregivers frequently face uncertainty when interpreting their child's health indicators — physical measurements, behavioural patterns, dietary reactions, vaccination status, and environmental risk factors — without immediate access to medical consultation. Existing digital health tools are either too generic (symptom checkers) or too technical (clinical dashboards) for everyday parenting use.

NurtureAI addresses this gap by providing a unified platform that distils complex multi-domain health data into understandable risk scores, predictions, and conversational guidance. The platform tackles four distinct ML prediction problems and one RAG-powered conversational module:

- Adult Behavior & Parenting Risk — lifestyle and behavioral risk scoring for parents/caregivers using the BRFSS 2013 dataset, including composite a Mental Stress Score, a Lifestyle Risk Score and a Physical Risk Score.
- Celiac Disease Screening — frequently underdiagnosed due to heterogeneous presentation; serological and symptomatic features (IgA/IgG/IgM) enable early screening.
- Infant Mortality Risk (<1 year) — a critical outcome in low-resource settings where socioeconomic, maternal, and vaccination factors are strong predictors (NFHS-5, 30 features).
- Older Child Behaviour & PIU — Problematic Internet Use risk for children aged 5–22, captured via the Child Mind Institute CMI-PIU dataset with 17 physical and psychometric predictors.
- Medical Q&A Chatbot (RAG) — 16,407 MedQuAD pairs indexed in ChromaDB with Gemma 4 31B for natural-language parenting queries with emergency triage flags.

## 1.2 Objectives

- Design a unified multi-model inference platform serving four ML prediction modules and one RAG chatbot through a single Flask REST API.
- Implement a wizard-style UX that progressively collects health data across parent and child profiles, minimising user burden while maximising data quality.
- Train and deploy a RandomForestClassifier on BRFSS 2013 data to predict adult parenting behavior risk with composite Mental Stress Score, a Lifestyle Risk Score and a Physical Risk Score.
- Train and deploy a GradientBoostingRegressor on CMI-PIU data to predict SII (0–3) with a composite behaviour_score (0–100).
- Train and deploy an L1-regularized Logistic Regression on NFHS-5 survey data (top 30 selected features) to predict infant mortality risk.
- Build a Celiac Disease binary classifier using symptom flags and serological markers (IgA/IgG/IgM) to flag at-risk children.
- Integrate a RAG-based medical chatbot powered by ChromaDB and Google Gemma 4 31B with emergency triage flags and multi-turn context.
- Deliver a responsive, mobile-first SPA with guided assessment flows, results visualisation, and conversational AI.

## 1.3 Key Features at a Glance

| Feature | Description | Priority |
|---|---|---|
| **Behaviour Risk** | GradientBoosting on CMI-PIU — SII (0–3) + behaviour_score (0–100) | Critical |
| **Child Mortality** | Random Forest on NFHS-5 — 30 socioeconomic + vaccination features | Critical |
| **Celiac Disease** | Binary classifier — symptoms + IgA/IgG/IgM serological markers | Critical |
| **AI Chatbot (RAG)** | ChromaDB 16,407 MedQuAD pairs + Gemma 4 31B + triage flags | Critical |
| **Adult Behavior Analysis** | RandomForest on BRFSS 2013 — composite mental stress, lifestyle and physical risk score | Critical |
| **Wizard UX** | Step-by-step guided data collection across parent + child profiles | Medium |
| **Flask REST API** | 5 endpoints + background thread loading + CORS + JSON responses | High |

# 2. Team Assignments & Dataset Overview

NurtureAI is developed by a team of five, with each member owning an end-to-end module from data preprocessing through API deployment — four members building independent ML prediction models and one building the RAG chatbot. The table below provides a consolidated view of responsibilities, datasets, and API endpoints:

| Member | Module | Dataset | Algorithm | API Endpoint |
|---|---|---|---|---|
| Mechapaia | Adult Behavior Analysis | BRFSS 2013 | RandomForestClassifier | /api/parent-assessment |
| Akhil | Celiac Disease Prediction | Clinical Celiac Dataset | RF + LogisticRegression | /api/celiac-check |
| Apoorv | Medical Q&A Chatbot (RAG) | MedQuAD (16,407 pairs) | ChromaDB + Gemma 4 31B | /api/chatbot |
| Arpita | Child Mortality Risk (NFHS) | NFHS-5 Survey | LR L1 + LLM Fallback | /api/child-infant |
| Vanshika | Behaviour Risk (CMI-PIU) | Child Mind Institute | GradientBoostingRegressor | /api/child-health |

## 2.1 Dataset Size and Integrity

| Dataset | Records | Processed | Features | Module |
|---|---|---|---|---|
| CMI-PIU (Behaviour) | ~3,900 | ~3,500 | 17 | Behaviour Risk |
| NFHS-5 (Child Mortality) | ~25,000 | ~22,000 | 30 | Child Mortality |
| Celiac Disease (Clinical) | ~1,200 | ~1,150 | 12 | Celiac Screening |
| MedQuAD (RAG Corpus) | 16,412 | 16,407 | N/A | AI Chatbot |
| BRFSS 2013 (Adult Behavior) | ~490,000 | Subset | 6+ key | Adult Behavior Analysis |

Failure rates across all datasets are within acceptable bounds for real-world observational data, primarily arising from missing values in self-reported variables. The MedQuAD corpus is processed at 100% fidelity as it is a curated Q&A dataset with no missing fields.

# 3. Experimental Pipeline & Methodology

The project follows a strictly ordered analytical pipeline to preserve statistical validity across all five prediction modules:

| Data Collection | Data Storage | Pre-Proces sing | Feature Engineering | Model Training | EDA (Python) | Flask API | Frontend SPA |
|---|---|---|---|---|---|---|---|

This separation ensures no leakage between observational behavioural data and engineered features, preserves observational integrity of clinical and demographic variables, enables reproducibility of EDA outputs across notebook environments, and maintains a consistent JSON contract between the Flask API and the frontend SPA.

# 4. Data Preprocessing & Feature Engineering

## 4.1 CMI-PIU Behaviour Dataset (Vanshika)

The primary preprocessing challenge with CMI-PIU data is class imbalance in the SII target variable (SII=0: ~38%, SII=1: ~32%, SII=2: ~19%, SII=3: ~11%). This is addressed using class_weight='balanced' in the GradientBoostingRegressor, which reweights training samples inversely proportional to class frequency.

The centrepiece of Vanshika's engineering contribution is the composite Behaviour_Score (0–100), computed as a weighted sum across five health dimensions:

- PAQ Score (Physical Activity): (PAQ / 4.0) * 30 points — highest weight, reflecting physical activity's primacy in behavioural health.
- Fitness Endurance: (Max_Stage / 14.0) * 20 points — normalised cardiorespiratory endurance stage.
- Sleep Quality (inverted): ((80 - SDS) / 60.0) * 20 points — high SDS = poor sleep = lower score.
- Body Fat % (inverted): ((50 - BIA_Fat) / 50.0) * 15 points — higher body fat = lower score.
- Resting Heart Rate (inverted): ((120 - HR) / 70.0) * 15 points — lower resting HR = better cardiovascular fitness.

The final score is clamped to [0, 100]: behavior_score = max(0.0, min(100.0, raw_score)). This single interpretable metric gives parents an immediate, intuitive understanding of their child's overall lifestyle health.

## 4.2 NFHS-5 Child Mortality Dataset (Arpita)

Given the high dimensionality of the NFHS-5 survey (hundreds of candidate features), Arpita implements automated feature selection using Logistic Regression with L1 regularization (Lasso) via the liblinear solver. The SelectFromModel class from scikit-learn retains the top 30 most predictive features, drastically reducing model complexity while preserving predictive signal.

This approach is particularly valuable for epidemiological datasets, as it produces interpretable, sparse models where each selected feature has a demonstrable statistical relationship with infant health outcomes. The 30 retained features span four categories: Household infrastructure (Toilet_Facility, Water_Source_Other, HealthInsurance), Maternal indicators (Resp_height, Tot_child_born, Curr_BrstFeed), Vaccination coverage (DPT_full, MEASLES_full, HepatitisB_atBirth, MMR, VitaminA), and Geographic one-hot state flags (Bihar, Uttar Pradesh, Uttarakhand, and others).

Input aliasing further improves API usability: delivery_place maps to DeliveryPlace_Private, water_source to Water_Source_Other, and state strings to the appropriate one-hot column — without affecting model integrity.

## 4.3 Celiac Disease Dataset (Akhil)

The Celiac Disease pipeline applies LabelEncoder to all categorical variables, including Gender, Diabetes (Yes/No), Diabetes Type (Type 1/2/Unknown), and the five binary symptom flags (Diarrhoea, Abdominal pain, Short_Stature, Sticky_Stool, Weight_loss). Missing data is handled via median imputation for continuous variables (IgA, IgG, IgM) and mode imputation for categorical columns.

EDA reveals that the Diarrhoea + Abdominal pain co-occurrence achieves the highest positive predictive value for Celiac positivity (PPV ≈ 0.71), while Short_Stature combined with Weight_loss is the second-highest discriminating pair (PPV ≈ 0.65). Type 1 Diabetes comorbidity elevates posterior Celiac probability by approximately 18% relative to non-diabetic patients.

## 4.4 MedQuAD RAG Corpus (Apoorv)

Each chunk is embedded using `all-MiniLM-L6-v2` (exported to ONNX for low-latency inference), and retrieved via cosine similarity. Metadata—such as focus area, and source IDs—is stored alongside embeddings in ChromaDB to support filtered and explainable retrieval.

At query time, the system constructs an enriched query by combining the user prompt with child-info, parent-info, and condensed chat-history, improving intent resolution. A top-k nearest-neighbour search retrieves relevant chunks. These rag-fetched documents are then fed into the Google's Gemma API and appropriate response is generated. This pipeline ensures context-aware, high-recall retrieval while maintaining real-time performance for multi-turn medical guidance.

## 4.5 BRFSS 2013 Adult Behavior Dataset (Mechapaia)

The BRFSS 2013 pipeline targets adult parenting behavior and associated lifestyle risks. Six key features drive the model: General_Health (1–5 Likert scale), Sleep_Hours (1–18), Exercise_Any (binary), Smoked_100_Cigs (binary), Income_Level (1–8), and Marital_Status (1–6). Beyond these raw inputs, Mechapaia engineers custom composite risk scores — a Mental Stress Score, a Lifestyle Risk Score and a Physical Risk Score — derived from combinations of behavioral variables to capture latent risk dimensions not directly measured in the raw survey.

A stratified train-test split ensures representative class distribution across training and validation folds, which is critical given health risk label imbalance in large-scale population surveys. The RandomForestClassifier (n_estimators=200, max_depth=12) is selected for its robustness to noisy, high-dimensional survey data and its ability to provide feature importance scores aligned with clinical interpretability requirements.

# 5. Model Selection, Implementation & Evaluation

Each prediction module uses an independently trained machine learning model, serialised as a .pkl file and served via the Flask API. The table below summarises the four ML models and the RAG chatbot:

| Module | Algorithm | Target Variable | Output Classes | Key Hyperparameters |
|---|---|---|---|---|
| Behaviour Risk (CMI-PIU) | GradientBoostingRegressor | SII (0-3) | 4 classes (None/Mild/Moderate/Severe) | class_weight='balanced' |
| Child Mortality (NFHS) | RandomForestClassifier | Mortality Risk (binary) | Low / Moderate / High | n_estimators=200, max_depth=12 |
| Celiac Disease | RF + LogisticRegression | Celiac +/- | Binary | LabelEncoder + 80-20 split |
| Adult Behavior Analysis (BRFSS) | RandomForestClassifier | Parenting Behavior Risk | Healthy / Moderate Risk / High Risk | n_estimators=200, max_depth=12, stratified split |
| AI Chatbot (RAG) | ChromaDB + Gemma 4 31B | Conversational Q&A | Open-ended + triage flags | all-MiniLM-L6-v2 ONNX, top-k retrieval |

## 5.1 Behaviour Risk — GradientBoostingRegressor (Vanshika)

Algorithm: GradientBoostingRegressor maps the 17 CMI-PIU physical and psychometric features to continuous SII severity values, which are subsequently mapped to four discrete Risk Labels (None, Mild, Moderate, Severe) using threshold-based classification.

Rationale: Gradient boosting is particularly well-suited to this problem because behavioural and lifestyle features exhibit non-linear, interaction-dependent relationships with internet use risk. The sequential tree-boosting approach captures these relationships more effectively than single-tree or linear methods. class_weight='balanced' addresses the SII class imbalance documented in EDA.

EDA Findings: Observed SII distribution: SII=0 (~38%), SII=1 (~32%), SII=2 (~19%), SII=3 (~11%). Median SDS score increases monotonically across SII categories (44, 49, 54, 60), confirming sleep disturbance as a reliable proxy indicator for PIU risk. Children in higher SII categories show elevated BMI, elevated waist circumference, and lower fitness endurance scores, consistent with a bidirectional relationship between physical inactivity and problematic internet use.

Evaluation: Performance is assessed using mean absolute error and R-squared on the regression output, supplemented by multi-class accuracy and per-class classification report on the discretised risk labels. Feature importance analysis confirms sleep quality and physical activity as the dominant predictors.

## 5.2 Child Mortality Risk — RandomForestClassifier (Arpita)

Algorithm: Two-stage pipeline: first, L1-regularized Logistic Regression identifies the top 30 NFHS-5 features via SelectFromModel; a final RandomForestClassifier (n_estimators=200, max_depth=12) is trained on these selected features. A Gemini LLM fallback generates natural-language explanations for edge-case predictions.

EDA Findings: Full DPT completion reduces mortality risk probability by approximately 28% in the dataset. MEASLES_full shows a similar protective effect. Vaccination completion is the dominant protective factor across all 30 features, confirmed by Random Forest Gini impurity analysis. Top 5 features by importance: DPT_full (0.142), MEASLES_full (0.118), Resp_height (0.096), Tot_child_born (0.087), VitaminA (0.075). Bihar and Uttar Pradesh show elevated baseline risk even after controlling for vaccination coverage.

Evaluation: Accuracy, F1-score for the high-risk class, AUC-ROC, and Gini feature importance are reported. Feature coefficient magnitudes are verified against established clinical risk factors for infant mortality.

## 5.3 Celiac Disease — RF & Logistic Regression (Akhil)

Algorithm: Both RandomForestClassifier and LogisticRegression are trained and evaluated on the preprocessed celiac dataset. The best-performing model on the validation set is selected for deployment. An 80-20 stratified split is maintained throughout.

EDA Findings: Celiac Positive patients exhibit significantly higher IgA and IgG levels ($p < 0.001$, Mann-Whitney U test) with clear distributional separation. IgA is the primary serological discriminator; IgG provides secondary discrimination. IgM contributes to the ensemble as part of the 12-feature input vector.

Evaluation: Accuracy, full classification reports (precision, recall, F1-score), and confusion matrices are generated for both models. Sensitivity on the celiac-confirmed class is prioritised, as clinical use cases demand high recall for true positive screening.

## 5.4 Adult Behavior Analysis — RandomForestClassifier (Mechapaia)

Algorithm: RandomForestClassifier with n_estimators=200 and max_depth=12 trained on BRFSS 2013 data. The model takes six behavioral inputs and outputs a three-class parenting risk prediction: Healthy, Moderate Risk, or High Risk. Per-class confidence probabilities are returned alongside the primary label, alongside composite Mental Stress, Lifestyle Risk and Physical Risk Score derived from the engineered features.

Rationale: Random forests are robust to noisy, high-dimensional survey data and handle mixed feature types natively. The model's hyperparameters (n_estimators=200, max_depth=12, min_samples_split=10) were explicitly selected to balance predictive power and generalization, preventing overfitting on the training data. Furthermore, feature importance scores provide clinically interpretable insight into which behavioral factors most strongly predict parenting risk.

Evaluation: Model performance is validated using a rigorous 80/20 stratified train-test split, ensuring representative class distributions across the data and confirming model stability. Overall accuracy, alongside per-class precision, recall, and F1-scores, are analyzed using a comprehensive classification report and a detailed confusion matrix. Sleep duration and exercise participation emerged as the top-ranked predictive features by Gini impurity decrease.

## 5.5 AI Medical Chatbot — RAG with Gemma 4 31B (Apoorv)

Architecture: ChromaDB cosine-similarity search retrieves the top-k most semantically similar MedQuAD passages for any parent query. These passages, together with the original query, child_info, and chat_history, are passed as structured context to Google Gemma 4 31B , which generates a parent-friendly, medically grounded response.

Emergency Triage: The structured LLM output includes isSerious (bool) and shouldCallEmergency (bool) flags. When triggered, the frontend SPA surfaces an emergency escalation prompt, ensuring life-threatening symptom descriptions receive appropriate urgency signalling.

Evaluation: Response quality is assessed using retrieval precision (relevance of retrieved passages) and answer faithfulness (alignment of generated response with retrieved context). Human evaluation across a sample of test queries confirms accurate, safe, and parent-friendly responses for common paediatric health questions.

We have also used a more powerful AI to analyse the response of the model we already use for quality testing of answers (LLM as a judge) , although it is not implemented in the frontend system.

# 6. System Architecture & Integration

## 6.1 Flask Application Entry Point (app.py)

The entry point initialises NurtureAI by creating the Flask application instance, enabling CORS, and performing asynchronous model loading via background daemon threads. The MODEL_REGISTRY dictionary maps logical module names to handler file paths; PKL_PATHS maps names to serialised .pkl model files. The _load_all_models_background() function spawns five daemon threads simultaneously, ensuring the server is immediately responsive while models initialise in parallel.

| Module Key | Handler File | API Endpoint |
|---|---|---|
| behaviour | behaviour_analysis/nurture_model.py | /api/child-health |
| child_mortality | child_mortality/child_health_model.py | /api/child-infant |
| adult_behavior (BRFSS) | nurture_model/nurture_model.py | /api/parent-assessment |
| celiac | celiac-disease/celiac_model.py | /api/celiac-check |
| chatbot | chatbot/chatbot_model.py | /api/chatbot |

## 6.2 Request Lifecycle — 4-Stage Inference Pipeline

Every user interaction passes through four ordered stages:

- Stage 1 — Flask Route Reception: request.get_json() validates body is not None; returns HTTP 400 if missing.
- Stage 2 — Model Status Check: _model_status[name].get('loaded') checked; returns HTTP 503 if not yet loaded or failed.
- Stage 3 — Module Dispatch: _modules[name].predict(data) called; returns structured result dict.
- Stage 4 — Response Serialisation: jsonify(result) returns HTTP 200 + JSON body to frontend SPA.

## 6.3 RAG Chatbot Pipeline

The RAG pipeline: the parent's query is embedded using all-MiniLM-L6-v2 (ONNX) and submitted to ChromaDB for cosine-similarity search against 16,412 MedQuAD embeddings. The top n_results documents are retrieved and injected as context into a structured Gemma 4 31B prompt alongside child_info and chat_history. The LLM generates a medically grounded, parent-friendly response. The isSerious and shouldCallEmergency boolean flags from the structured LLM output trigger emergency escalation prompts in the frontend UI.

## 6.4 Frontend SPA Architecture

The NurtureAI frontend (index.html, style.css, app.js) is a Single Page Application built with vanilla HTML5, CSS3, and JavaScript ES6+, served directly by Flask. It communicates exclusively via the REST API using fetch(). The UI implements a warm, calming aesthetic suited to a parenting health companion, deliberately contrasting with clinical dashboard interfaces.

- Screen 1 — Welcome: NurtureAI branding, tagline, 'Get Started' CTA. Under 3 minutes to complete.
- Screen 2 — Parent Setup Wizard (4 steps): Name, health profile (sleep, exercise, smoking, income, marital status), child profile (age, sex, BMI, height, weight, waist, BP, heart rate, SDS, fitness, BIA), and additional metrics.
- Screen 3 — Assessment Results: Colour-coded risk cards for all applicable models, behaviour_score progress bar, risk label badges, and plain-language recommendations.
- Screen 4 — Chatbot Interface: Conversational UI with chat history, typing indicators, child_info context injection, and emergency response flag rendering.

## 6.5 Technology Stack

| Library | Version | Role in NurtureAI |
|---|---|---|
| Flask | >=3.0.0 | Web framework — REST API routing and static SPA serving. |
| Flask-CORS | >=4.0.0 | Cross-origin resource sharing for frontend-backend communication. |
| scikit-learn | >=1.2.0 | Random Forest, Logistic Regression, GradientBoosting, SelectFromModel, LabelEncoder. |
| pandas / numpy | >=2.0.0 | Data manipulation, feature DataFrame construction, composite score computation. |
| chromadb | >=0.4.0 | Vector store for RAG — 16,412 MedQuAD Q&A embeddings (chroma_full_state.pkl). |
| google-genai (Gemini) | >=0.1.0 | Gemma 4 31B API for chatbot response generation and infant health LLM fallback. |
| all-MiniLM-L6-v2 (ONNX) | HuggingFace | Sentence embedding model for ChromaDB vector indexing and query embedding. |
| importlib / threading / pickle | stdlib | Dynamic module loading (MODEL_REGISTRY), background daemon threads, model serialization. |

# 7. Exploratory Data Analysis (EDA)

EDA was conducted using Python (Jupyter Notebooks) to understand distributional, relational, and temporal properties of data across all primary modules. Statistical findings directly informed preprocessing decisions, feature engineering priorities, and model hyperparameter choices.

## 7.1 CMI-PIU Behaviour Dataset — EDA Findings

### 7.1.1 SII Distribution

Observed distribution: SII=0 (None, ~38%), SII=1 (Mild, ~32%), SII=2 (Moderate, ~19%), SII=3 (Severe, ~11%). The combined Moderate+Severe population (~30%) represents a clinically significant group requiring attention. Median(SII) < Mean(SII) confirms right skew in risk severity distribution, consistent with stochastic risk process behaviour. Class imbalance was addressed using class_weight='balanced' in the GradientBoostingRegressor.

### 7.1.2 Physical Health vs. Behavioural Risk

Children in higher SII categories (Moderate/Severe) show elevated BMI and waist circumference, lower fitness endurance stage scores, and shorter endurance times. This pattern suggests a bidirectional relationship between physical inactivity and problematic internet use behaviour, consistent with established adolescent health literature.

### 7.1.3 Sleep Disturbance Analysis

A monotonically increasing median SDS score is observed across SII=0 through SII=3 (median SDS ≈ 44, 49, 54, 60 respectively). Observed correlation: corr(SDS, SII) > 0. Higher sleep disturbance scores are strongly correlated with higher SII categories, indicating that sleep disruption is a reliable proxy indicator for problematic internet use risk.

### 7.1.4 Physical Activity Patterns

Lower PAQ-A and PAQ-C scores correlate with higher SII in both adolescent and child cohorts. The effect is more pronounced for PAQ-A (teens aged 14–22), suggesting that reducing structured physical activity during adolescence is a risk amplifier for problematic internet use.

## 7.2 NFHS-5 Child Mortality Dataset — EDA Findings

### 7.2.1 Vaccination Coverage Analysis

Full DPT completion reduces mortality risk probability by approximately 28% in the dataset. MEASLES_full shows a similar protective effect. Vaccination completion is the dominant protective factor across all 30 features, confirmed by Random Forest Gini feature importance (DPT_full: 0.142, MEASLES_full: 0.118, VitaminA: 0.075, HepatitisB_atBirth: 0.061).

### 7.2.2 Social Determinant Interactions

Households with toilet facilities and health insurance show substantially lower child mortality risk across all states. The interaction between State (geographic location) and social determinants explains significant variance. Bihar and Uttar Pradesh show elevated baseline risk even after controlling for vaccination coverage, underscoring the role of geographic socioeconomic disparities in child health outcomes.

## 7.3 Celiac Disease Dataset — EDA Findings

### 7.3.1 Symptom Co-occurrence Analysis

A co-occurrence matrix computed for the seven binary symptom features reveals: Diarrhoea + Abdominal pain co-occurrence achieves PPV ≈ 0.71 for Celiac positivity; Short_Stature + Weight_loss is the second-highest discriminating pair (PPV ≈ 0.65); Type 1 Diabetes comorbidity elevates posterior Celiac probability by approximately 18% relative to non-diabetic patients.

### 7.3.2 Serological Marker Distribution

Celiac Positive patients exhibit significantly higher IgA and IgG levels with clear distributional separation ($p < 0.001$, Mann-Whitney U test). IgA is the primary serological discriminator; IgG provides secondary discrimination. IgM shows less discriminative power individually but contributes to the ensemble model within the 12-feature input vector.

# 8. Limitations

- Cross-sectional data: The CMI-PIU and NFHS-5 datasets are single-point-in-time observations; longitudinal dynamics of child health are not captured.
- Demographic skew: CMI-PIU participants are primarily aged 5–22 from English-speaking contexts; cross-cultural generalisability is limited.
- Model interpretability: Random Forest and Gradient Boosting models are ensemble methods; per-prediction explanations require additional tools (e.g., SHAP).
- Chatbot knowledge boundary: The ChromaDB vector store is limited to the embedded MedQuAD corpus; queries outside this domain may yield generic responses.
- Self-reported data: Physical activity (PAQ-A/PAQ-C) and some NFHS-5 variables are self-reported, introducing measurement bias.
- Hardware constraint: All models were trained and served locally on consumer hardware; scalability to large concurrent user loads has not been tested.

# 9. Future Scope

- Multi-user longitudinal data collection: Extend CMI-PIU analysis to track behavioural risk trajectories over time for individual children, enabling early intervention.
- Model explainability: Integrate SHAP (SHapley Additive exPlanations) values to provide per-prediction feature attribution for clinical interpretability.
- MLOps pipeline: Deploy MLflow/Airflow for periodic model retraining on updated survey data, maintaining accuracy as population health patterns evolve.
- Expanded chatbot corpus: Ingest peer-reviewed paediatric guidelines, WHO child health resources, and UNICEF reports into ChromaDB to broaden chatbot coverage.
- Cloud deployment: Migrate Flask to AWS/GCP for concurrent multi-user access and automatic model scaling.
- Real-time wearable integration: Incorporate biometric feeds (heart rate, sleep tracking) into the behaviour model for continuous risk monitoring.
- Personalised health timelines: Longitudinal tracking of individual user inputs to generate personalised health trajectories and proactive risk alerts.
- Federated learning: Privacy-preserving model training across distributed healthcare institutions without centralising sensitive patient data.

# 10. Conclusion

NurtureAI successfully delivers a robust and practical multi-model child health intelligence platform, effectively bridging clinical-grade machine learning with consumer-friendly parenting tooling. By combining four independently trained ML models — covering adult behavior and parenting risk, celiac disease screening, infant mortality risk, and older child behavioural/PIU assessment — alongside a RAG-powered medical chatbot, all served within a single Flask SPA, the project demonstrates both technical depth and practical utility.

From a statistical standpoint, child health outcomes across all three core datasets exhibit structures consistent with stochastic behavioural processes: right-skewed SII distributions in CMI-PIU data, session-dependent vaccination-mortality correlations in NFHS-5 records, and symptom co-occurrence clustering in celiac data.

Key strengths include the modular microservice-inspired architecture with clean module separation, the RAG chatbot backed by 16,412 real medical Q&A pairs with emergency triage flags, the wizard-style UX making clinical data collection accessible to non-technical parents, the NFHS-5 grounding in India-specific epidemiological evidence, and Mechapaia's composite Mental Stress, a Lifestyle Risk and Physical Risk Score feature engineering on the BRFSS dataset demonstrating domain-adapted ML methodology.

In essence, NurtureAI is not just a project — it is a solid foundation for a scalable, production-ready child health companion platform that could be extended with longitudinal tracking, wearable integration, and multilingual support for broader deployment across diverse parenting populations.

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